

Estimation of apple orchard yield using night time imaging

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Abstract

This work investigates the potential of night time imaging for estimating apple orchard yield. Forty two trees were photographed from two sides with cameras mounted at three heights. Each image was analyzed and the results of the six images associated with each tree were summed up to provide a ‘tree count’. Fourteen trees were selected randomly in order to calibrate a relationship between the ‘tree count’ estimate and the actual tree yield. This relationship was then applied to the ‘tree count’ results of the remaining trees. Although the yield estimate error for a single tree was sometimes large, the overall yield estimate was within 10% of the actual yield.

Keywords: artificial vision, fruit detection, image processing

Introduction

Automated estimation of the expected yield is still a major challenge in orchards. Currently, yield estimation is based on visual inspection of some trees by the farmer, which is both time consuming and prone to errors. At first glance, it may seem that this task could be easily automated using computer vision. However, the unstructured and cluttered environment of the tree canopy makes this task very challenging, especially when the fruits are featureless and with a color similar to that of the canopy (e.g. Kurtulmus *et al.*, 2011; Linker *et al.*, 2012; Payne *et al.*, 2013; Zhou *et al.*, 2012; Bansal *et al.*, 2013; Kelman and Linker, 2014). Furthermore, uneven daylight illumination complicates the task and this has led a number of researchers to investigate the use of night time imaging (e.g. Sites and Delwiche, 1988; Payne *et al.*, 2014; Cohen *et al.*, 2014; Linker and Kelman, 2015). Encouraging results were reported in the recent studies of Cohen *et al.* (2014) and Linker and Kelman (2015), which both focused on the detection of green mature apples in night time images. Both studies reported strong linear relationships between the number of objects identified as apples and the number of apples counted by visual inspection of the images. However, in neither study, was an attempt made to correlate the results of the image analysis with the actual yield at the tree level. In the present study, both methods were applied to a dataset containing over 250 pictures of 42 trees and the suitability of each approach for estimating tree yield was investigated.

Materials and methods

Data collection

The imaging system consisted of three identical cameras (Canon PowerShot G7. Resolution 3,648×2,736 pixels) mounted on a vertical pole at 110, 180 and 240 cm from the ground and two additional poles positioned at 60 cm from the camera pole, on which LED lights were mounted at heights of 130 and 250 cm. The whole assembly was mounted on a trailer which was driven in the middle of the tree rows. Each tree was photographed from two sides, resulting in six images per tree. The imaging system was used to record images of 42 Golden Delicious trees at the Matityahu Research Station in Northern Israel on July 17, 2014. The trees were picked individually at the end of the season in order to obtain ground truth yield data. In addition, the number of apples visible in each image was determined by visual inspection of the images.

Data analysis

The images were processed with two methods. The first method (labeled Approach #1 below), described in Linker *et al.* (2012) and Cohen *et al.* (2014), uses color-and-texture analysis followed by comparison of the contours of the obtained blobs with a simple model (circle). This method includes four main steps: (1) pixel classification based on color and texture; (2) detection of blobs of connected pixels and extension of these blobs to neighboring pixels with similar properties; (3) segmentation of the contours of the blobs into arcs; and (4) grouping of arcs into objects and comparison with the reference model (circle). The second method (Approach #2), described in Linker and Kelman (2015), relies on analysis of the spatial distribution of light around specular highlights (bright spots) that occur (mostly but not only) on apples when artificial illumination is used. This procedure is based on the observation that although such highlights also occur on leaves, because of the spherical shape of apples, the light distribution around highlights located on apples is such that pixels with intensity above some threshold form a circle or part of a circle. Accordingly, detection of apples is achieved via three main steps: (1) identification of a local maximum of the light intensity and extraction of a sub-image centered at this point; (2) formation of regions of connected pixels (so-called light patches) corresponding to decreasing light intensities; and (3) analysis of the geometrical properties of the sequence of patches obtained with decreasing intensity thresholds. Although both methods are expected to detect the same objects, detection is achieved via completely different features of the 'apple' object: in Approach #1, detection relies primarily on pixel color and texture and on the assumption that the circular contour on the apple is at least partially visible. In Approach #2, color is not considered at all and apples are detected solely via their shape, assuming that the visible region of the apple is sufficiently large to enable analysis of the light distribution on its surface.

Both methods were calibrated using 12 images (four images at each height) selected randomly. The calibrated procedures were applied to all the images and, for each approach, the number of objects detected in the six images of a tree were summed up to produce a 'tree count'. Fourteen trees were selected randomly in order to derive a relationship between the 'tree counts' and the actual tree yield. This relationship was then applied to all the 'tree count' results to produce an estimate of the yield of each tree. In order to investigate the influence of the choice of the trees used for calibration on the results, this last step was repeated ten times, each time with a different subset of trees for calibration.

Results

Calibration

The parameter values reported in Linker *et al.* (2012) (Dataset#1) and Linker and Kelman (2015) were used as default values for the respective algorithms and the number of parameter adjustments was kept at a minimum.

Approach #1 includes a color-and-texture classifier which is not generic since the apparent color of the apples depends not only on the apple variety and their developmental stage but also on the illumination. Accordingly, about half of the apples were marked manually in the 12 calibration images (the remaining apples were masked out), creating 'apple' and 'non-apple' regions from which random pixels were sampled to train the classifier. The only parameters which were adjusted were the ones related to the apple size: minimum radius: 30 pixels, expected radius: 60 pixels, and maximum radius 150 pixels. Figure 1 shows the detection results obtained with these values for a typical image. Figure 2 (left) summarizes the results for the 12 calibration images and shows the relationship which was obtained between the number of detected objects and the number of apples according to visual inspection. This relationship was used to convert the number of detected objects into the estimated number of apples in all 252 images and the results are shown in the right frame of Figure 3. It can be seen that there is a good agreement between the two.



Figure 1. Detection results of Approach #1 for a typical image (middle camera). The detected objects are indicated by circles superimposed on the original image.

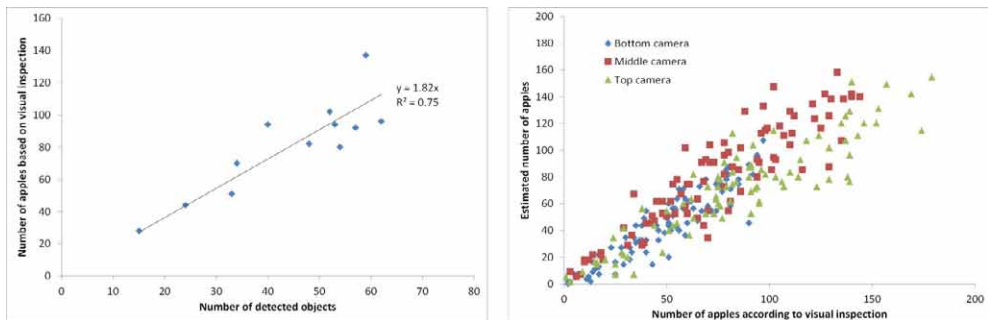


Figure 2. Approach #1. Left: Relationship between the number of detected objects and the number of apples according to visual inspection in the 12 calibration images. Right: Number of apples according to visual inspection vs automatic estimation using the linear model shown in the left frame (all 252 images).



Figure 3. Detection results of Approach #2 for the image shown in Figure 1. The centers of the detected objects are indicated by '+' superimposed on the original image.

Approach #2 includes far fewer parameters than Approach #1 and most importantly does not use color information, which makes it much more robust with respect to changes in the conditions in which the images are acquired (illumination and camera type). Indeed, since the size range of the apples was similar to that of the 2013 dataset used by Linker and Kelman (2015), the corresponding set of parameters was used without any change. Figure 3 shows the detection results for the same image as in Figure 2, and Figure 4 (left) summarizes the results for the 12 calibration images. As with Approach #1, a linear relationship between the number of detected objects and the number of apples according to visual inspection was obtained, and this relationship was used to convert the number of detected objects into the estimated number of apples in all 252 images. The results are shown in Figure 4 (right). Comparing this Figure with Figures 2 (right) shows that the results of the two approaches appear to be quite similar. Table 1 confirms that this is indeed the case and although the two approaches are based on completely different properties of the apples (color-and-texture vs reflectance of spherical object), their overall performance is very similar. Overall both approaches gave total estimates which were within 6% of the visual count. The worst results were obtained for the images of the top camera analyzed using Approach #1. These poorer results can be explained

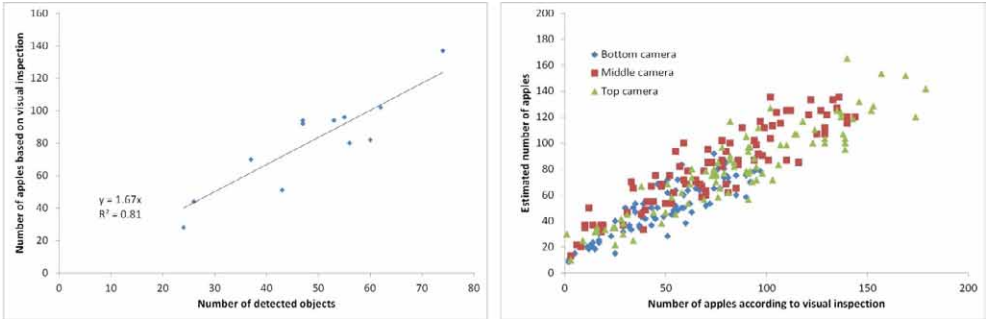


Figure 4. Approach #2. Left: relationship between the number of detected objects and the number of apples according to visual inspection in the 12 calibration images. Right: number of apples according to visual inspection vs automatic estimation using the linear model shown in the left frame (all 252 images).

Table 1. Comparison between the number of apples according to visual inspection and automatic estimation.

	Bottom images	Middle images	Top images	All images
Number of apples according to visual count	4,218	6,551	7,473	18,242
Approach #1				
Number of apples estimated by automatic procedure	3,871	6,887	6,334	17,191
Actual/estimated ratio	0.92	1.07	0.85	0.94
Approach #2				
Number of apples estimated by automatic procedure	4,362	7,209	7,092	18,663
Actual/estimated ratio	1.03	1.10	0.95	1.02

by the poor quality of the illumination in the upper part of the tree in some of the images, which resulted in many apples being too dark to be recognized by the color classifier.

Yield estimation

As mentioned above, the number of objects detected in the six images of a tree were summed up to produce a 'tree count' and fourteen trees were selected randomly in order to derive a relationship between the 'tree counts' and the actual tree yield. This relationship was then used to produce an estimate of the yield of each tree. This analysis was repeated 10 times using different trees for calibration.

Typical results (i.e. those corresponding to Models A1 and B1, Figure 5 and 6) obtained with Approaches #1 and #2 are shown in Figure 5 and 6, respectively. The left frames show the relationship between the 'tree counts' and the actual yield of the calibration trees. The right frames show the actual yield vs the estimated yield obtained using the relationship shown in the corresponding left frame. The left frames show that, regardless of the method used to analyze the images, the number of objects detected in the six images of a tree corresponds to only about one third of the actual tree yield. Still, the right frames show that this simple model enables estimation of the yield of the

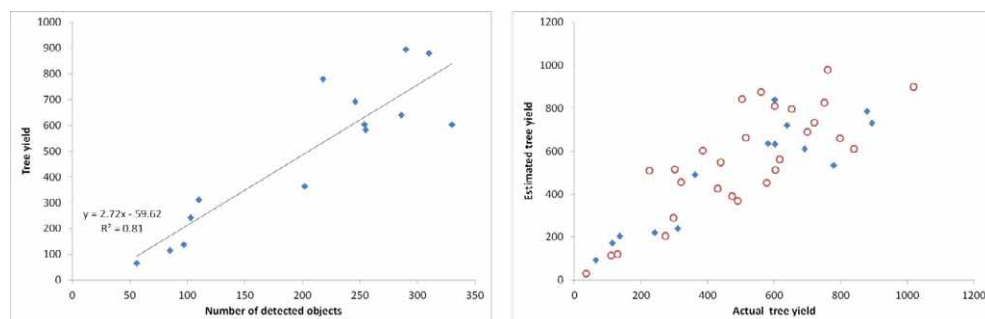


Figure 5. Yield estimation results using Approach #1 (Model A1). Left: Relationship between the number of objects detected in the six images associated with a tree and the actual tree yield. Right: actual yield vs yield estimated using the model shown in the left frame. The filled symbols correspond to the calibration trees and the open symbols correspond to the validation trees.

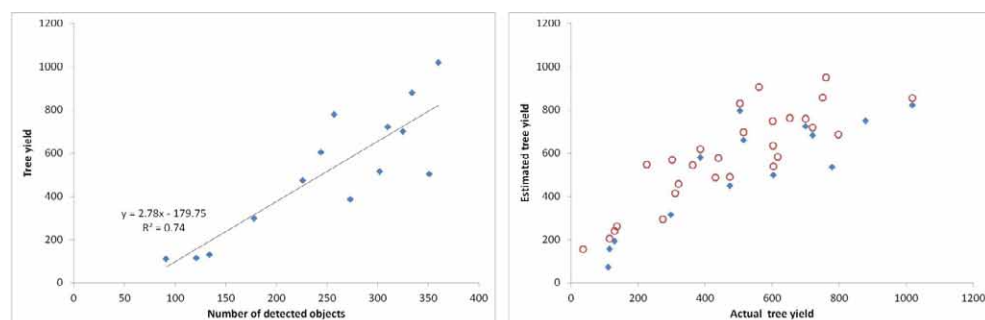


Figure 6. Yield estimation results using Approach #2 (Model B1). Left: Relationship between the number of objects detected in the six images associated with a tree and the actual tree yield. Right: actual yield vs yield estimated using the model shown in the left frame. The filled symbols correspond to the calibration trees and the open symbols correspond to the validation trees.

remaining trees with reasonable accuracy. Although the yield estimation error for a single tree is sometimes large (up to ~300 apples in extreme cases), the first rows of Tables 2 and 3 show that the overall yield estimate is within 10% of the actual yield. Furthermore, the additional rows in these tables show that the overall performance is not affected strongly by the choice of the trees used for calibration, and the error exceeds 10% only once for each approach.

Conclusions

Night time imaging appears to be a promising technique for estimating apple yield, at least at the row scale. The two approaches investigated in this study led to overall estimates which were within 10% of the actual yield using less than 15 trees for calibration. Although these results need to be confirmed using a larger dataset, one can envision a self-calibrating system which would require the farmer to provide fruit counts for 15-20 trees. This information would be used to establish the

Table 2. Approach #1. Yield estimations based on ten random sub-sets of 14 calibration trees.

Model name	Validation trees only			All trees (yield: 21,047)	
	Actual yield	Estimated yield	Error (%)	Estimated yield	Error (%)
A1	14,240	15,321	7.59	22,128	5.14
A2	13,917	12,619	-9.33	19,749	-6.17
A3	13,596	14,410	5.98	21,861	3.87
A4	14,210	14,528	2.24	21,365	1.51
A5	13,726	13,231	-3.60	20,552	-2.35
A6	13,280	12,612	-5.03	20,379	-3.17
A7	12,878	14,085	9.37	22,254	5.73
A8	14,076	14,274	1.41	21,245	0.94
A9	13,989	13,552	-3.12	20,610	-2.08
A10	14,725	13,180	-10.49	19,502	-7.34

Table 3. Approach #2. Yield estimations based on ten random sub-sets of 14 calibration trees.

Model name	Validation trees only			All trees (yield: 21,047)	
	Actual yield	Estimated yield	Error (%)	Estimated yield	Error (%)
B1	13,812	15,190	9.98	22,425	6.55
B2	13,792	14,721	6.74	21,976	4.42
B3	13,576	14,917	9.88	22,388	6.37
B4	14,360	14,064	-2.06	20,751	-1.41
B5	13,276	13,448	1.30	21,219	0.82
B6	13,459	14,761	9.67	22,349	6.18
B7	13,941	13,288	-4.69	20,394	-3.10
B8	14,250	14,366	0.81	21,163	0.55
B9	13,644	13,703	0.44	21,106	0.28
B10	14,018	15,771	12.51	22,800	8.33

relationship between detections and yield (i.e. left frames of Figure 5 and 6), based on which the system would produce the overall yield estimate. The key issue is the calibration of the algorithm for identifying the apples in the images. In this respect, Approach #2 has the advantage that it does not use color information and hence does not require training a dataset-specific classifier. Although the training of the classifier itself is straightforward, it requires careful labeling of apple regions in the calibration images, which is time consuming.

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